Assignment 1

Question (1):-

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[n-k]\, h[k]$$

$n=0$: $y = h[0]$

$n=1$: $y = 3h[0] + h[1]$

$n=2$: $y = 2h[0] + 3h[1] + h[2]$

$n=3$: $y = -h[0] + 2h[1] + 3h[2] + h[3]$

$n=4$: $y = -h[1] + 2h[2] + 3h[3] + h[4]$

$n=5$: $y = -h[2] + 2h[3] + 3h[4] + h[5]$

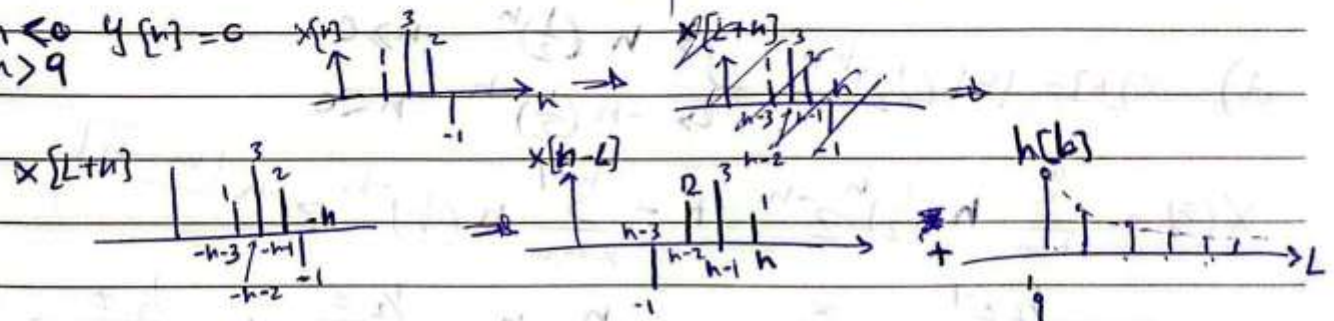
$n=6$: $y = -h[3] + 2h[4] + 3h[5] + h[6]$

$n=7$: $y = -h[4] + 2h[5] + 3h[6]$

$n=8$: $y = -h[5] + 2h[6]$

$n=9$: $y = -h[6]$

$n<0$, $n>9$ $y[n] = 0$



Question (2):-

a) $x[n] = 2^n u[n] + 3\left(\frac{1}{2}\right)^n u[n]$

$$X(z) = \sum_{n=-\infty}^{\infty} 2^n u[n] z^{-n} + 3\sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n u[n] z^{-n}$$

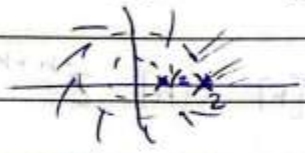
$$= \sum_{0}^{\infty} 2^n z^{-n} + 3\sum_{0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n}$$

$$\frac{1}{1-2z^{-1}} + \frac{3}{1-\frac{1}{2}z^{-1}}$$

$|2z^{-1}| < 1$, $|z| > 2$

$\left|\frac{1}{2}z^{-1}\right| < 1$, $|z| > \frac{1}{2}$

ROC: $|z| > 2$



b) $x[n] = \left(\frac{1}{2}\right)^n u[n+1] + 3^n u[-n-1]$

$$< \sum_{n=-1}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} + \sum_{n=-\infty}^{-1} 3^n z^{-n} \quad m=-n \quad = 2z + \sum_{n=0}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} + \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n z^n$$

$|z| > \frac{1}{2}$

$$= 2z + \frac{1}{1-\frac{1}{2}z^{-1}} + \sum_{n=0}^{\infty} \left(\frac{1}{3}\right)^n z^n - 1 = 2z + \frac{1}{1-\frac{1}{2}z^{-1}} + \frac{1}{1-\frac{z}{3}} - 1$$

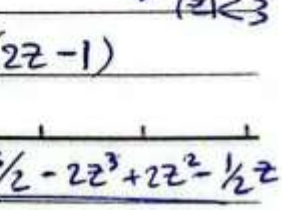
$|z| < 3$

$$= 2z - 1 + \frac{z}{z-\frac{1}{2}} + \frac{3}{3-z} = \frac{3z - z^2 + 3z - \frac{3}{2}}{(z-\frac{1}{2})(3-z)} + (2z-1)$$

$$= \frac{z^2 - 6z + \frac{3}{2} + (2z^2 - z - z + \frac{1}{2})(3-z)}{(z-\frac{1}{2})(z-3)} = \frac{z^2 - 6z + \frac{3}{2} + 6z^2 - 6z + \frac{3}{2} - 2z^3 + 2z^2 - \frac{1}{2}z}{(z-\frac{1}{2})(z-3)}$$

$$= \frac{-2z^3 + 9z^2 - 12z + 3}{(z-\frac{1}{2})(z-3)}$$

ROC: $\frac{1}{2} < |z| < 3$



c) $x[n] = \left(\frac{1}{3}\right)^n \sin\left(\frac{\pi n}{4}\right) u[n] = \left(\frac{1}{3}\right)^n \frac{1}{2j}\left(e^{j\frac{\pi n}{4}} - e^{-j\frac{\pi n}{4}}\right) u[n]$

$$X(z) = \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n e^{j\frac{\pi n}{4}} z^{-n} - \frac{1}{2j}\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n e^{-j\frac{\pi n}{4}} z^{-n}$$

$|z| > 1/3$

$$= \frac{1}{2j}\left[\frac{1}{1-\frac{1}{3}e^{j\pi/4}z^{-1}} - \frac{1}{1-\frac{1}{3}e^{-j\pi/4}z^{-1}}\right] \rightarrow |z| > \frac{1}{3}$$

$$= \frac{1}{2j}\left[\frac{z}{z-\frac{e^{j\pi/4}}{3}} - \frac{z}{z-\frac{1}{3e^{j\pi/4}}}\right] = \frac{1}{2j}\left[\frac{z^2 - \frac{z}{3e^{j\pi/4}} - z^2 + z\frac{e^{j\pi/4}}{3}}{\left(z-\frac{e^{j\pi/4}}{3}\right)\left(z-\frac{1}{3e^{j\pi/4}}\right)}\right]$$

ROC $|z| > \frac{1}{3}$

d) $x[n] = |n|\left(\frac{1}{2}\right)^{|n|} = \begin{cases} n\left(\frac{1}{2}\right)^n & n \geq 0 \\ -n\left(\frac{1}{2}\right)^{-n} & n < 0 \end{cases}$

$$X(z) = \sum_{n=0}^{\infty} n\left(\frac{1}{2}\right)^n z^{-n} + \sum_{n=-\infty}^{n=-1} n\left(\frac{1}{2}\right)^{-n} z^{-n} \quad (m = -n)$$

$$= \frac{\frac{1}{2}z^{-1}}{\left(1-\frac{1}{2}z^{-1}\right)^2} + \sum_{n=1}^{\infty} n\left(\frac{1}{2}\right)^n z^n = \frac{\frac{1}{2}z}{\left(z-\frac{1}{2}\right)^2} + \frac{\frac{1}{2}z}{\left(1-\frac{1}{2}z\right)^2}$$

$|z| > 1/2$ ← → $|z| < 2$

ROC $\frac{1}{2} < |z| < 2$

Question (3) :-

$$r_{xx}(l) = \sum_n x[n]\,x[n-l]$$

$$R_{xx}(z) = \sum_{L=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x[n]\,x[n-l]\,z^{-L} \Rightarrow \quad n - L = k$$

$$= \sum_{k=-\infty}^{\infty}\sum_{n=-\infty}^{\infty} x[n]\,x[k]\,z^{-n+k} = \sum_{k=-\infty}^{\infty} x[k]\,z^{k} \sum_{n=-\infty}^{\infty} x[n]\,z^{-n}$$

$X(z^{-1})$ $X(z)$

$= X(z)\,X(z^{-1})$ ROC ROC $x(z) \cap$ ROC $x(z^{-1})$

$\alpha < |z| < \beta \cap \frac{1}{\alpha} > |z| > \frac{1}{\beta}$

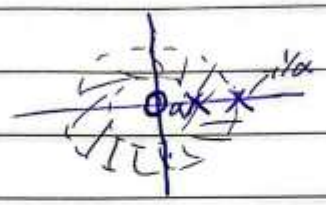
ROC $\max\left(\frac{1}{\beta}, \alpha\right) < |z| < \min\left(\beta, \frac{1}{\alpha}\right)$

b) $x[n] = \alpha^n u[n] \quad |\alpha| < 1$

$X(z) = \frac{1}{1-\alpha z^{-1}}$, ROC : $|z| > |\alpha|$ $\rightarrow \frac{z}{z-\alpha}$

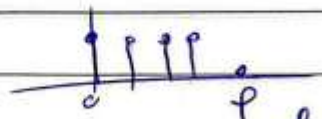
$X(z^{-1}) = \frac{1}{1-\alpha z}$, ROC $|z| > 1/\alpha$

So, $R_{xx}(z) = \frac{z}{(z-\alpha)(1-\alpha z)}$



ROC $\quad \alpha < |z| < \frac{1}{\alpha}$

c) $r_{xx}[l] = \sum_{n=-\infty}^{\infty} \alpha^n u[n]\, \alpha^{n-l} u[n-l]$



$= \sum_{n=0}^{l-1} \alpha^n \alpha^n \alpha^{-l} = \alpha^{-l} \sum_{n=0}^{l-1} (\alpha^2)^n = \alpha^{-l} \frac{1-(\alpha^2)^l}{1-\alpha^2}$

$= \frac{\alpha^{-l} - \alpha^l}{1-\alpha^2}$

Question (1e):-

$x[n] = -\frac{1}{3}\left(\frac{1}{2}\right)^n u[n] - \frac{4}{3} 2^n u[-n-1]$

$X(z) = \frac{-1}{3} \sum_{n=-\infty}^{\infty} \left(\frac{1}{2}\right)^n z^{-n} u[n] - \frac{4}{3} \sum_{n=-\infty}^{\infty} 2^n z^{-n} u[-n-1]$

$= -\frac{1}{3} \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{4}{3} \left(\sum_{n=-\infty}^{-1} 2^n z^{-n} \right)$ → $\left(\sum_{1}^{\infty} 2^{-n} z^{n} \right)$ $\left(\sum_{0}^{\infty} \left(\frac{1}{2}\right)^n z^n - 1 \right)$

$= -\frac{1}{3} \frac{1}{1-\frac{1}{2}z^{-1}} - \frac{4}{3} \frac{1}{1-\frac{1}{2}z} - 1$

$-\frac{1}{3}\left(\frac{1}{1-\frac{1}{2}z^{-1}} + 4 \frac{-2z^{-1}}{-2z^{-1}+1} \right) - 1 = \frac{-4 + \frac{35}{2}z^{-1} - 7z^{-2}}{3(1-\frac{1}{2}z^{-1})(1-2z^{-1})}$

$Y(z) = X(z) H(z) \Rightarrow 1 - z^{-2} = \left(\frac{-4}{3} + \frac{35}{6}z^{-1} - \frac{7}{3}z^{-2}\right) H(z)$

$H(z) = \frac{1-z^{-2}}{\frac{-4}{3} + \frac{35}{6}z^{-1} - \frac{7}{3}z^{-2}}$

Question (5):

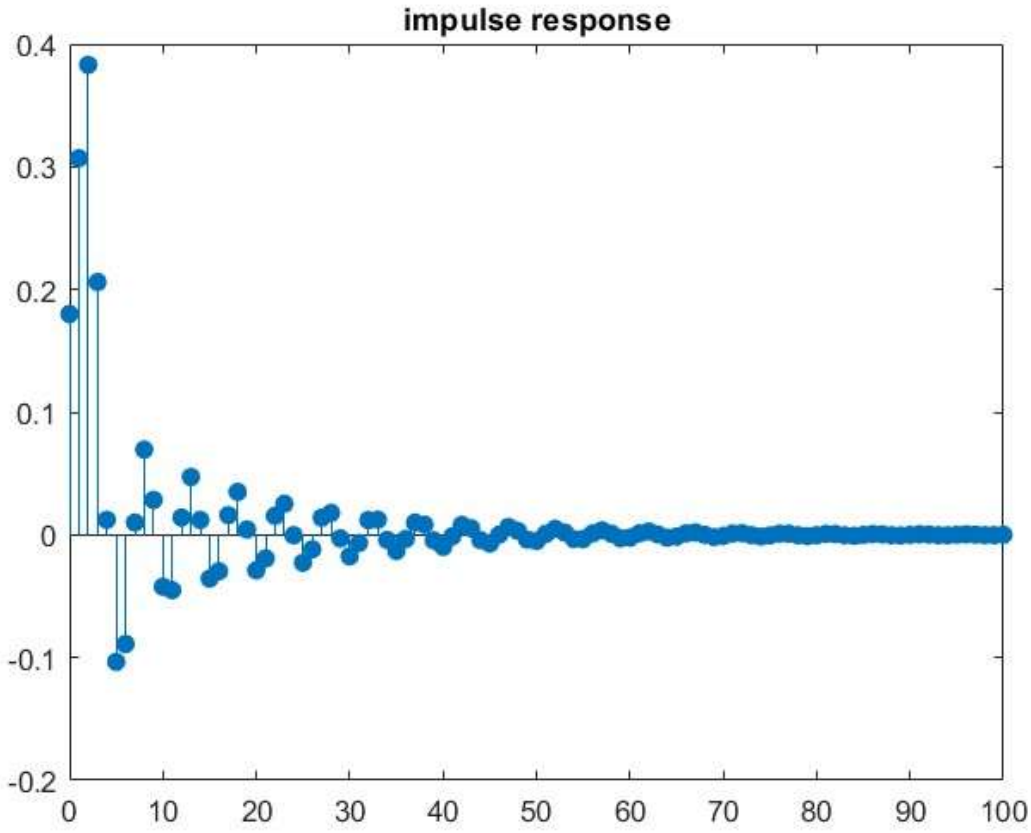
Setup:

```
clear;
close all;
clc;

b=[0.18,0.1,0.3,0.1,0.18];
a=[1,-1.15,1.5,-0.7,0.25];
n=[0:100];
```

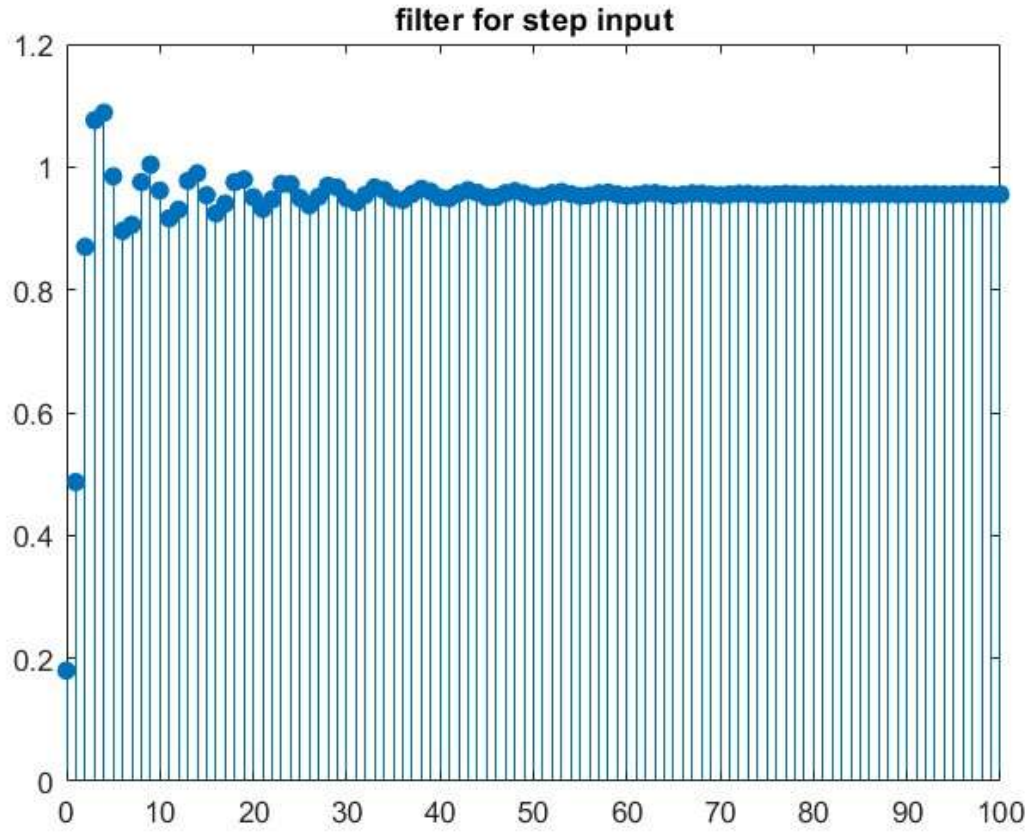
(a)

```
%%impulse response
H_impulse=impz(b,a,101);
figure;
stem(n,H_impulse,"filled");
title("impulse response")
```



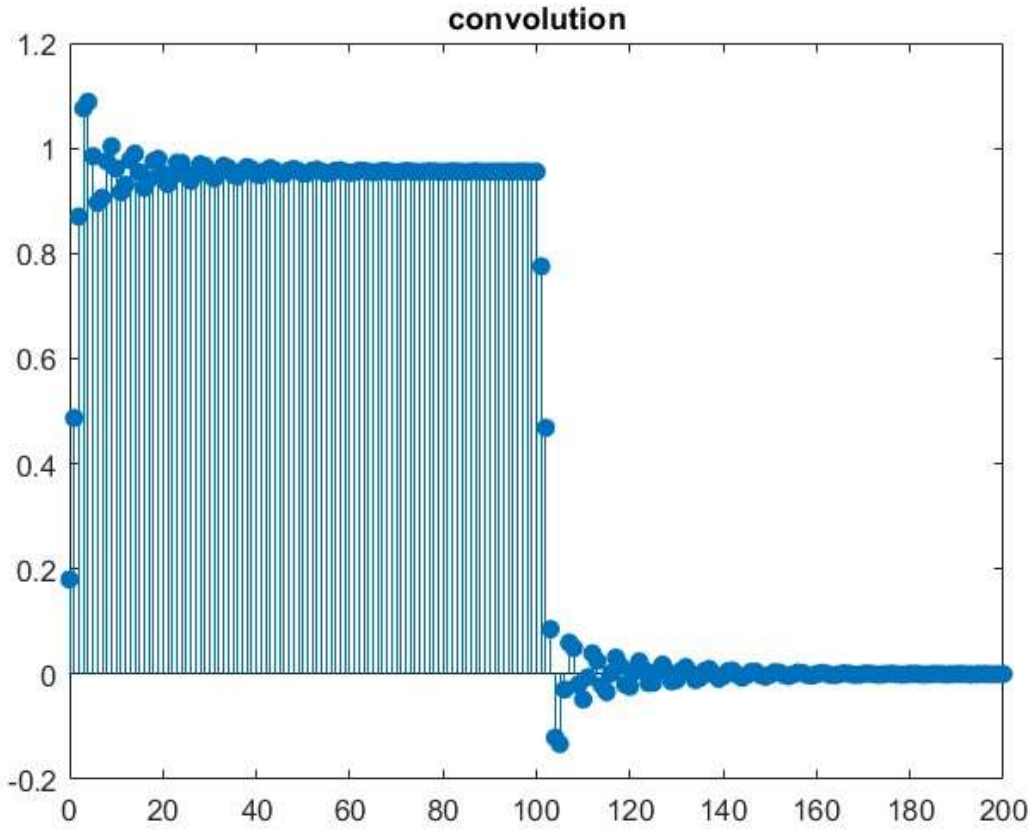
(b)

```
%%step response
u_step=ones(101,1);
y_step=filter(b,a,u_step);
figure;
stem(n,y_step,"filled");
title('filter for step input')
```



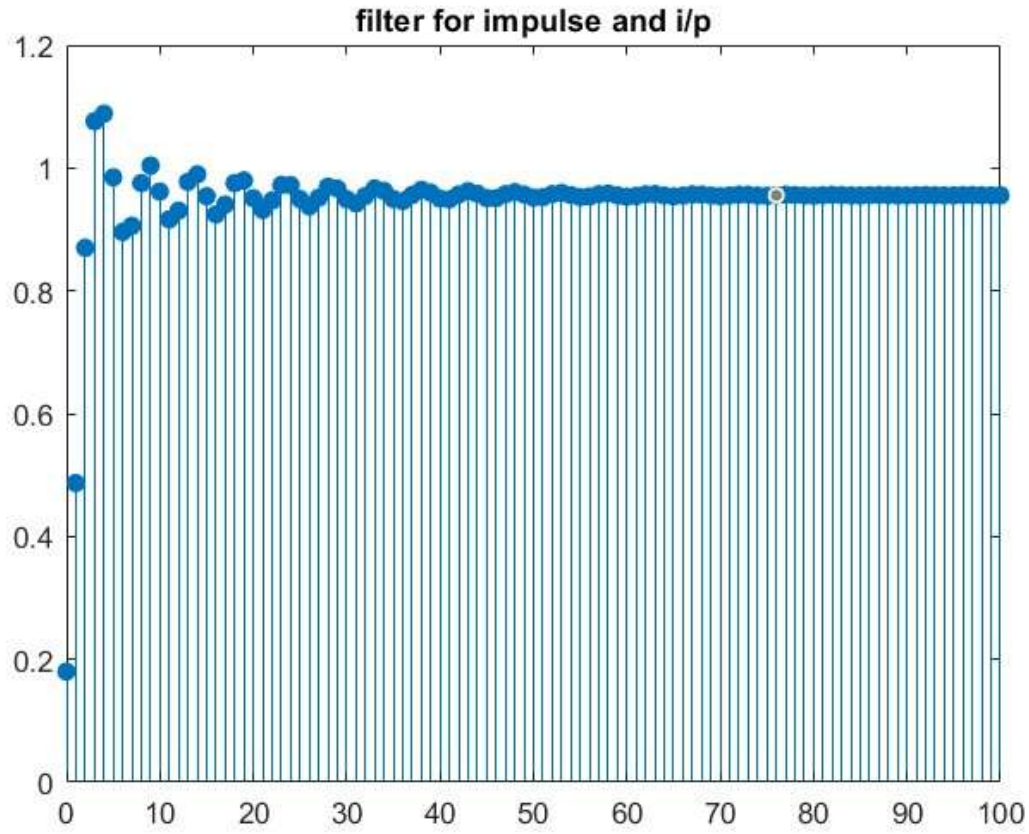
(c)

```
%%convolution
y_conv=conv(H_impulse,u_step);
figure;
n_conv=[0:200];
stem(n_conv,y_conv,"filled");
title("convolution")
```



(d)

```
%%filter
y_filter=filter(H_impulse,1,u_step);
figure;
stem(n,y_filter,"filled");
title("filter for impulse and i/p")
```



Notes

(a) Is the impulse response of our system
(b) Is the response of iir system the depends on previous feedback when we introduce a step input
(c) A convolution between the impulse response and the step input which should get us the same result as (b)
(d) Filter(h,1,u) is considering an fir system with no feedback and a step input on the impulse response system